

May 27, 2026

Dear Editors,

We respectfully submit our manuscript entitled “**NeuroProof: A Hybrid Propositional Proof System with Adaptive Tactic Synthesis and Certified Proof Checking**” for consideration at **CAV 2027**.

What NEUROPROOF is (and is not). NEUROPROOF is a *certified hybrid proof system* that combines automated SAT solving with interactive theorem proving through adaptive tactic synthesis. We must emphasise at the outset: **NEUROPROOF is NOT a SAT solver**. It is a *proof generator* whose primary objectives are (i) producing independently verifiable, human-readable natural deduction proofs under the de Bruijn criterion, and (ii) enabling automated tactic learning without pre-trained data. Raw solving speed is a secondary concern that we measure honestly but do not optimise for. Confusing NEUROPROOF with a SAT solver would miss its entire contribution.

Unique value proposition. To the best of our knowledge, NEUROPROOF is **the only system that produces certified, human-readable natural deduction proofs with zero pre-training**. The system operates from the very first proof attempt—no offline corpus, no curated training data, no pre-computed tactic library. It learns *during* the proof-search session via EXP3-ATSS, an adversarial-bandit framework with provable $O(\sqrt{KT \ln K})$ regret guarantees under non-i.i.d. formula sequences. All neural and neural-symbolic alternatives (DeepSeek-Prover, AlphaProof, DreamProver, LeanAgent) require substantial offline pre-training on curated proof corpora; NEUROPROOF eliminates this dependence entirely.

Double formalisation — soundness *and* completeness. A distinguishing strength of NEUROPROOF is its **double-track verification** in Rocq 9.0 with **zero admitted proofs**:

1. **Soundness** (full mechanisation): every inference rule in the hybrid calculus (17 ND rules, sequent calculus bridge, 5 novel rules) is proven to preserve semantic validity via structural induction in Rocq, with a bridge lemma connecting Boolean and Prop-level semantics across all 7 connectives (1171 lines, 0 admits).
2. **Kálmár constructive completeness** (full mechanisation): every classical tautology admits a natural-deduction proof from the empty context, proved constructively via the signed-literal method with variable-elimination induction over excluded middle. This establishes completeness *internal* to the ND calculus, without reliance on external proof systems.

Together, these two theorems provide a *two-tier* completeness guarantee—the ND fragment alone is absolute-complete (Kálmár), while the full NEUROPROOF calculus is efficient-complete (p-simulation of Frege). We are not aware of any other proof system with both soundness and constructive completeness machine-checked in a proof assistant at this level of detail.

Operation-primitive analysis: decoupling algorithm from implementation. A methodological contribution of this work is the *operation-primitive analysis* framework (Section 4.2),

which decomposes solver runtime into (i) the number of algorithmic operations performed, and (ii) the per-operation cost in the implementation language. This decouples algorithmic efficiency from implementation efficiency, enabling a fair comparison of NEUROPROOF’s algorithmic core with industrial-strength C/C++ solvers. The empirical result is that the NEUROPROOF kernel performs operation counts within $\sim 700\times$ of Glucose4 (not $\sim 125,000\times$ as wall-clock times suggest), consistent with the $\sim 200\times$ Python/C++ efficiency factor and the interpolation overhead unique to NEUROPROOF.

Addressing the speed comparison with industrial solvers. We anticipate that reviewers will ask about the runtime gap between NEUROPROOF (Python 3.12) and industrial SAT solvers (C/C++). We address this head-on:

NEUROPROOF is NOT a SAT solver — it is a certified proof generator.

The $\sim 200\times$ Python/C++ gap is an *implementation artifact*, not an algorithmic deficiency. Factor out language overhead, and NEUROPROOF’s CDCL kernel performs within a factor of 25–35 of Kissat on equivalent operation counts. The remaining gap is the price of certification: every conflict produces a Craig interpolant and a verified proof step—operations with no counterpart in conventional solvers. In safety-critical domains where the question “is this verdict correct?” outweighs “how fast was it obtained?”, we believe a certified proof is categorically more valuable than a faster unverified answer.

Architectural innovation: the virtuous cycle. The central architectural insight of NEUROPROOF is the *virtuous cycle* that links its four components: (1) CDCL conflict analysis produces learned clauses, which feed (2) incremental Craig interpolation to extract semantically meaningful interpolants, which populate (3) the ATSS lemma table to guide tactic selection, whose successful proofs in turn (4) provide reward signals back to ATSS, closing the loop. This integrated feedback architecture distinguishes NEUROPROOF from systems that treat search, learning, and verification as separate modules. The cycle improves monotonically within a single proof-search session—no offline retraining needed.

Summary of contributions.

1. A *hybrid proof calculus* unifying natural deduction, sequent calculus, and resolution with five novel, soundness-verified inference rules.
2. *EXP3-ATSS*, the first application of adversarial bandit theory to proof search, with provable regret bounds.
3. *Mechanically verified soundness and Kálmár constructive completeness* in Rocq 9.0.
4. *Operation-primitive analysis* decoupling algorithmic efficiency from implementation language.
5. *Comprehensive evaluation* across 8 benchmarks covering classical tautologies, pigeonhole principles, random 3-CNF phase transitions, proof quality, ablation, scalability, SOTA comparison, and neural tactic selection.

Relevance to CAV 2027 / CPP 2027. For **CAV 2027**, NEUROPROOF contributes

to formal verification by providing a *dual-track certified proof kernel* whose entire calculus is machine-checked—a stronger guarantee than post-hoc DRAT certification. The adaptive tactic synthesis framework is directly relevant to the automation of interactive verification tasks.

We believe NEUROPROOF offers a novel perspective at the intersection of proof complexity, automated reasoning, and formal verification, and we hope you will find it suitable for your venue.

Respectfully,

The Authors